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Functional Disorders of the Lower Urinary Tract in Children

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Keywords

child; daytime incontinence; nocturnal enuresis; urinary bladder; urinary bladder/physiology; urinary incontinence; urination disorders

Questions

1. A 9-year-old girl presents to the office with a chief complaint of daytime urinary incontinence. She denies dysuria, hematuria, or enuresis. She has never had a urinary tract infection (UTI) and there is no history of hydronephrosis. Her mother states that toilet training occurred at 2 years of age and was "easy." A clean-catch urinalysis has been obtained by her pediatrician and is completely normal. When characterizing the incontinence, she states that she completely soaks her clothes, necessitating a change in wardrobe. She denies urgency or frequency and otherwise voids every 2 to 3 hours during the day. She states that the episodes are often associated with laughing at a funny joke or movie. She denies any issues with constipation. Assuming physical examination and voiding diary are normal, which of the following would be a reasonable first-line treatment strategy?
    a. Acupuncture
    b. Biofeedback

c. Hypnosis
d. Imipramine
e. Oxybutynin

2. A 4-year-old girl presents to the office with a chief complaint of labial adhesions, dysuria, daytime urinary incontinence, and recurrent UTIs. Her mother states that toilet training was completed at 20 months and that there were no issues with this. Based on history and completed voiding diary, her elimination pattern is normal. Her mother states that during the past 3 months, however, she has noted that her underpants are damp and that this is often noted within a few minutes of the child having voided. They deny any urgency or frequency. She recently has been complaining of severe dysuria with two urinalyses (UAs) in the past 4 weeks demonstrating 5 to 10 white blood cells per high-power field (WBCs/hpf). Urine cultures have all been negative, and antibiotics have not helped with symptomatology. Physical examination reveals superficial labial adhesions with moderately erythematous external genitalia. Noninvasive urodynamics with pelvic ultrasound and uroflowmetry reveal a bell-shaped curve and an empty bladder with normal wall thickness. There is a small amount of fluid noted in the vagina. While the child climbs down off of the examining room table, there is involuntary leakage of urine. Repeat pelvic ultrasound reveals that the vaginal vault is empty. What is the most likely diagnosis?
    a. Bladder-bowel dysfunction
    b. Dysfunctional voiding
    c. Urge urinary incontinence
    d. Vaginal reflux
    e. Vesicoureteral reflux (VUR)
3. An 8-year-old boy is brought in by his grandmother, who recently obtained guardianship due to parental divorce and the biological mother's recent untimely death. She states that his teachers have been complaining that for the last 4 months he spends most of his time in the restroom, asking to use the bathroom approximately every 15 minutes. He has had some occasional dysuria, but no hematuria or fevers. They deny any UTIs or daytime urinary

incontinence. He has occasional constipation, but this is readily corrected with fruit juice and he generally has one soft, smooth bowel movement daily. He is generally able to sleep through the night without having to wake to void, and they deny enuresis. The element in the history that will most often be able to distinguish pollakiuria from overactive bladder (OAB) is:

a. a recent life event.
b. male versus female gender.
c. no history of UTIs.
d. no urinary incontinence.
e. the child does not wake to void.

4. Lower urinary tract (LUT) dysfunction is associated with which of the following?

a. Constipation
b. Neuropsychiatric issues
c. UTIs
d. VUR
e. All of the above

5. The purported mechanism of action for botulinum toxin in the treatment of children and adolescents with dysfunctional voiding is:

a. reducing the frequency and intensity of uninhibited detrusor contractions during the filling phase of the bladder.
b. smooth muscle relaxation at the bladder neck.
c. paralysis of striated muscle of the external sphincter.
d. stabilization of the motor end plate, inhibiting spinal cord feedback loops.
e. none of the above.

6. A 7-year-old girl presents to your office with a 6-month history of recurrent UTIs, daytime urinary incontinence, urgency, dysuria, and enuresis. Mother states that her daughter will have wet underwear and clothes 3 to 4 times per week. She is also wet 5 out of 7 nights per week. Mother states that she will often “wait until the last minute” to void. There are no febrile UTIs, and urine cultures from her pediatrician’s office reveal multiple organisms on 2 separate occasions in the last 3 months. A renal/bladder ultrasound demonstrates normal upper urinary tracts. Urinalysis in the office is

negative and uroflow shows staccato waveform. You initiate behavioral modification and she returns 6 weeks later with improvement but remains wet during the day 1 time per week with urinary urgency and wets the bed 2 nights per week. The best next step is:

a. anticholinergics.
b. moisture alarm.
c. alpha blockers.
d. biofeedback.
e. botulinum toxin.

7. Which of the following organ systems is implicated in the pathogenesis of enuresis?

a. Bladder
b. Brain
c. Kidney
d. All of the above
e. None of the above

8. An 8-year-old boy presents to your office with his parents for consultation regarding treatment for primary nocturnal enuresis. Behavioral modification, desmopressin, and the enuresis alarm have failed. Which of the following parameters is the best predictor of response to treatment with desmopressin?

a. Age of child
b. Bladder capacity
c. Motivation of family
d. Nocturnal polyuria
e. Poor arousal

Answers

1 **b. Biofeedback. Giggle incontinence (enuresis risoria) is an uncommon form of daytime incontinence and is classically seen in school-aged females. Typically, there is moderate to large amounts of urinary leakage triggered by laughing alone.** The incontinence episodes are invariably significant, and often the

entire bladder volume is drained. Daytime urinary incontinence in conjunction with laughter is also seen in children with OAB and is more common than true giggle incontinence. It is a diagnosis of exclusion and is usually established on history and is supplemented by the absence of other voiding symptoms and normal investigations. Currently, available treatment strategies include biofeedback or methylphenidate.

2 **d. Vaginal reflux. Vaginal reflux (vaginal entrapment, vaginal voiding) is characterized by incontinence following normal voiding in the absence of other LUT symptoms. It is commonly seen in prepubertal girls, and the typical history is that of wetting of undergarments approximately 10 to 15 minutes following a normal void. It can often be associated with labial adhesions because of chronic irritation and inflammation from skin exposure to relatively caustic urine.** Reassurance and postural modification to ensure complete vaginal emptying is the only treatment that is required.

3 **e. The child does not wake to void. Pollakiuria is a disorder characterized by a very high daytime frequency of micturition (sometimes as high as 50 times per day). A key aspect of this syndrome, which differentiates it from OAB and can often clinch the diagnosis, is that the symptoms are limited to the daytime.** It is seen in early childhood (4 to 6 years of age) in both genders and associated with a history of recent death or life-threatening event in the family. **Usually, it runs a benign, self-limited course during a period of approximately 6 months.**

4 **e. All of the above. There are long-standing, clear associations between LUT dysfunction and bowel dysfunction, UTIs, VUR, and various psychiatric diagnoses.** The incomplete bladder emptying that occurs in children with LUT dysfunction can lead to urinary stasis, with subsequent UTIs causing inflammatory changes in the bladder wall that stimulate hypertrophy and overactivity. It has been theorized that detrusor hypertrophy can alter the closure mechanism at the ureterovesical junction (UVJ), leading to reflux. It has also been shown that ongoing issues with bowel-bladder dysfunction can have a negative effect on VUR resolution rates,

and that addressing bowel dysfunction alone can positively influence LUT function. Finally, clinicians should be cognizant of the association between neuropsychiatric diagnoses and daytime wetting, as the former is likely to interfere with treatment success of the latter.

5 **c. Paralysis of striated muscle of the external sphincter.** Botulinum-A toxin acts by inhibiting acetylcholine (ACh) release at the presynaptic neuromuscular junction. Inhibited ACh release results in regionally decreased muscle contractility and atrophy at the injection site, which in the case of dysfunctional voiding would be the striated muscle of the external urinary sphincter. The chemical denervation that ensues is a reversible process, and eventually the toxin is inactivated and removed. Clinical effects begin within 5 to 7 days of injection, with maximal effects reached within 4 to 6 weeks. The duration of induced paralysis varies depending on the type of muscle treated, with duration of treatment effect lasting between 3 and 12 months.

6 **d. Biofeedback.** All patients with lower urinary tract dysfunction (LUTD) should receive behavioral modification as treatment. A uroflow showing a staccato waveform along with incomplete emptying is consistent with dysfunctional voiding. **Biofeedback is particularly valuable in the treatment of dysfunctional voiding and improves LUTD by coordinating the pelvic floor muscle activity and abdominal muscle activity to promote efficient bladder emptying.**

7 **d. All of the above.** The three organ systems implicated in the pathogenesis of enuresis include the bladder (reduced nocturnal bladder capacity), the kidney (nocturnal polyuria), and the brain (e.g., a disorder affecting arousal from sleep). **Enuresis is logically thought to result from a disruption or maturational lag in one or more of these critical domains.**

8 **d. Nocturnal polyuria.** The nocturnal enuresis alarm and desmopressin are both valid treatment options. There exist patient, caregiver, and disease-related parameters that may aid in offering prognostic information in terms of which therapeutic modality should be first entertained. **The enuresis alarm seems best fit for**

motivated patients and families and for children without polyuria but with low voided volume. Desmopressin seems best suited for children with nocturnal polyuria and normal bladder reservoir function, for those with infrequent wet episodes, and for families in whom alarm treatment has failed or who have refused alarm treatment.

Chapter Review

1. Immature detrusor sphincter coordination manifested as detrusor hypercontractility and interrupted voiding commonly occurs in the first 2 years of life and results in functional bladder outflow obstruction.
2. Even in newborns, micturition does not occur during sleep, suggesting modulation of micturition by higher centers.
3. The association of constipation with urologic pathology is referred to as *bladder bowel dysfunction or BBD.* Abnormalities of bowel function are commonly present in young children with voiding dysfunction.
4. Giggle incontinence often results in complete emptying of the bladder.
5. In patients who develop acquired bladder sphincter dysfunction, a significant proportion also have bowel dysfunction.
6. There is a significant association of bladder dysfunction with nonresolution of high-grade vesicoureteral reflux.
7. Nocturnal urine output in many enuretic children is in excess of bladder reservoir capacity during sleep.
8. Many enuretic children have a marked reduction in functional bladder capacity when compared with age-matched controls and may have detrusor instability as well.
9. Overactive bladder is the most common lower urinary tract disorder in children, with a peak incidence between 5 and 7 years.
10. Behavioral and emotional disorders occur in 20% to 30% of children with lower urinary tract disorders.

11. Vaginal reflux (vaginal entrapment, vaginal voiding) is characterized by incontinence following normal voiding in the absence of other LUT symptoms. It is commonly seen in prepubertal girls, and the typical history is that of wetting of undergarments about 10 to 15 minutes following a normal void. It can often be associated with labial adhesions.
12. Pollakiuria is a disorder characterized by a very high daytime frequency of micturition (sometimes as high as 50 times per day). A key aspect of this syndrome, which differentiates it from OAB and can often clinch the diagnosis, is that the symptoms are limited to the daytime. It is seen in early childhood (4 to 6 years of age) in both genders and can be associated with stressors such as a history of recent death or life-threatening event in the family. Usually, it runs a benign, self-limited course over a period of approximately 6 months.
13. All children with LUTD should receive behavioral modification as treatment. Behavioral modification includes demystification, education, counseling, timed voiding, relaxation techniques, monitoring of stool consistency, and diary registration of bladder and bowel emptying patterns.
14. The use of an alarm in the treatment of enuresis seems best fit for motivated patients and families and for children without polyuria but with low voided volume. Desmopressin seems best suited for children with nocturnal polyuria and normal bladder reservoir function, for those with infrequent wet episodes, and for families for whom alarm treatment has failed or who have refused alarm treatment.

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Management of Defecation Disorders

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Keywords

bowel function; bowel management; defecation disorder; urinary function

Questions

1. The Rome IV criteria evaluate the following aspects of bowel function EXCEPT:
    a. pain with bowel movements.
    b. production of large stools that can block the toilet.
    c. number of bowel movements per week.
    d. duration of symptoms.
    e. all of the above.
2. A 4-year-old girl presents with a 6-month history of urinary frequency and urgency. Parents report daily small bowel movements, which they attribute to poor diet (“picky eater”). Her physical exam is normal. There is no history of urinary tract infections. Which of the following interventions can lead to paradoxical worsening of her symptoms?
    a. Increase in fluid intake
    b. Polyethylene glycol (PEG)
    c. Oxybutynin

d. Trimethoprim prophylaxis
e. Biofeedback

3. Which one of the following aspects of a patient's history and physical examination should raise suspicion for an underlying organic pathology causing constipation?
    a. Early age of onset (before toilet training)
    b. Presence of a palpable mass in the left lower quadrant
    c. Onset after diet change
    d. Poor dietary habits ("picky eater")
    e. All of the above
4. Which part of the physical exam can be safely omitted during initial evaluation of a child with suspected functional constipation?
    a. Height and weight
    b. Inspection of the lower back
    c. Lower extremity muscle tone and reflexes
    d. Digital rectal exam
    e. Visual inspection of the perineum
5. Which of the following metabolic/endocrinologic pathologies is unlikely to cause constipation?
    a. Hypercalcemia
    b. Hypokalemia
    c. Hypothyroidism
    d. Diabetes insipidus
    e. Precocious puberty
6. Regarding the surgical management of refractory constipation, which of the following statements is TRUE?
    a. Access for antegrade irrigations should be limited to the cecum.
    b. Open surgical intervention carries a higher success rate than laparoscopic or percutaneous procedures.
    c. Malone antegrade continence enema (MACE) channels and C-tubes provide better procedural independence than retrograde enemas for patients with neuropathic bowel dysfunction.
    d. Similar success rates can be expected irrespective of the underlying pathology.
    e. All of the above.

7. The main principle behind daily antegrade enemas for continence is:
    a. washout with regular evacuation of the entire colon.
    b. direct softening of stools to facilitate passage during the day.
    c. improve hydration.
    d. decrease colon motility.
    e. decrease sphincter tone.
8. Initial workup of a child with constipation is most likely to benefit from:
    a. abdominal radiograph.
    b. anal manometry.
    c. colonic transit time studies.
    d. magnetic resonance imaging (MRI) of lumbosacral spine.
    e. contrast enema study.
9. Which of the following statements regarding creation of a MACE channel is TRUE?
    a. Previous surgical interventions are a contraindication for a laparoscopic approach.
    b. An aggressive bowel washout and mechanical preparation are always warranted before surgery.
    c. Presence of a ventriculoperitoneal shunt is a contraindication for laparoscopic approach.
    d. The appendix may be of sufficient length to be split in order to create a MACE and Mitrofanoff channel for neuropathic bowel and bladder management.
    e. An anti-reflux mechanism (cecal wrap) is always required in order to prevent stool leakage.
10. Which of the following statements regarding cecostomy tubes is most accurate?
    a. It is a good alternative for patients who have previously undergone an appendectomy.
    b. It avoids the need for regular instrumentation.
    c. It is difficult to remove or convert to a MACE channel.
    d. The most common problem is stenosis and difficulty accessing for fluid instillation.
    e. It is a great alternative for families who have problems with compliance.

11. Recommendations for antegrade enema regimens should include which of the following?
    a. Sterile saline is preferred versus tap water or “homemade” saline solution.
    b. Daily enemas are universally required to achieve continence.
    c. Early morning irrigations are preferable as it allows the patients to enjoy better daytime continence.
    d. Trial and error for more than 6 months may be warranted to reach a reliable enema routine.
    e. All of the above.
12. What is the proposed mechanism of action of prucalopride?
    a. Stool softener
    b. Antispasmodic
    c. Bulking agent
    d. Prokinetic
    e. Antiinflammatory
13. Which of the following agents is preferred (first-line) medication for maintenance management of constipation?
    a. Milk of magnesia
    b. Mineral oil
    c. PEG
    d. Prucalopride
    e. Psyllium husk
14. Features of functional constipation include:
    a. recurrence despite recommendations consistent with optimal medical management.
    b. alternating constipation and diarrhea.
    c. episodes of bowel obstruction.
    d. bilious vomiting.
    e. ribbon-like stools.
15. Which one of the following stool characteristics is NOT included in the Bristol scale?
    a. Consistency
    b. Shape
    c. Difficulty having bowel movements
    d. Odor

e. None of the above

16. Of the following, which one is the best diagnostic test to confirm Hirschsprung disease?
    a. Barium enema
    b. Computed tomography (CT) scan with oral and intravenous contrast
    c. Rectal biopsy
    d. Colonoscopy
    e. Lumbosacral spine MRI
17. New onset of abdominal pain and distention after antegrade instillation of fluid in a child with a cecostomy tube in place for more than 6 months should alert the physician about:
    a. spontaneous cecal perforation.
    b. use of hypotonic fluid for irrigations.
    c. presence of a large fecal load in the rectum and descending colon.
    d. an incompetent ileocecal valve with retrograde irrigation into the terminal ileum.
    e. irritable bowel syndrome.
18. Which of the following statements regarding abdominal plain film during initial assessment of a child with functional constipation is FALSE?
    a. Amount and distribution of fecal material can predict likelihood of response and recurrence with medical therapy
    b. Helps assess for fecal impaction
    c. Helps assess response to bowel washout
    d. Aids in demonstrating for parents or caretakers the presence of constipation
    e. Detects bony abnormalities that may be associated with neuropathic bowel dysfunction
19. Which statement regarding diagnosis and management of functional constipation is TRUE?
    a. Assessment during evaluation of a child with recurrent urinary tract infections is warranted.
    b. It is an integral part of a program dealing with dysfunctional voiding.

c. It should be addressed before proceeding with surgical interventions for vesicoureteral reflux.
d. It can be managed by pediatric urologists and urology nurse practitioners.
e. All of the above.

Imaging

1. The abdominal radiograph (left) and pelvic ultrasound (right) shown in Fig. 36.1 were obtained on a 5-year-old girl with recurrent abdominal pain and distention for the past year. She has a palpable soft, nontender mass in the left lower quadrant. Which one of the following interventions is LEAST likely to help with initial management?
   a. Digital disimpaction under sedation
   b. Enemas
   c. "High-dose" PEG
   d. Supplement fiber intake
   e. Increase fluid intake

Answers

1. **e. All of the above. The Rome IV criteria requires symptoms to be present for at least 1 month, and takes into account developmental age and absence of an underlying organic pathology.** Pain with defecation, history of large-diameter stools that may obstruct the toilet, and infrequent bowel movements (≤2 defecations per week) are all included in the diagnosis of functional constipation.
2. **c. Oxybutynin.** Constipation is a common side effect of medications used to deal with lower urinary tract symptoms (particularly antimuscarinics). Because of this common association, during the past few decades pediatric urologists have become comfortable with assessing and managing bowel problems. **Development or worsening of constipation will sometimes**

coincide or worsen with precipitating factors, such as change in diet (for example, transitioning out of breastfeeding) and introduction of new medications (such as oxybutynin for management of urinary frequency). Increase in fecal load in the rectum, coupled with toilet avoidance due to lower abdominal pain and discomfort with defecation, may lead to paradoxical worsening of lower urinary tract symptoms. Increase in fluid intake and use of PEG is unlikely to worsen lower urinary tract symptoms and is part of initial recommendations for bowel retraining. Similarly, biofeedback may have a neutral or beneficial effect. Antibiotic prophylaxis, in the absence of recurrent infections, is not expected to influence symptomatology.

3. **a. Early age of onset (before toilet training). The age of onset of symptoms is one of the easiest and most important pieces of information to obtain, as it can be an important indicator for underlying pathology, particularly if symptoms have been present since early in life (infancy).** Other critical information to be actively gathered includes failure to toilet train within an age-appropriate and developmental timeframe, pain with defecation, bleeding per rectum, associated abdominal pain, fecal incontinence, holding behaviors, nausea or vomiting, weight loss, growth pattern (including height and weight), developmental delay, and failure to thrive. Patients with functional constipation often present or worsen after dietary changes. Not uncommonly, otherwise healthy children are described by parents as very selective (or “picky”). In the setting of significant fecal load or impaction in a child with constipation, a palpable mass in the left lower quadrant (“fecaloma”) may be detected during examination.

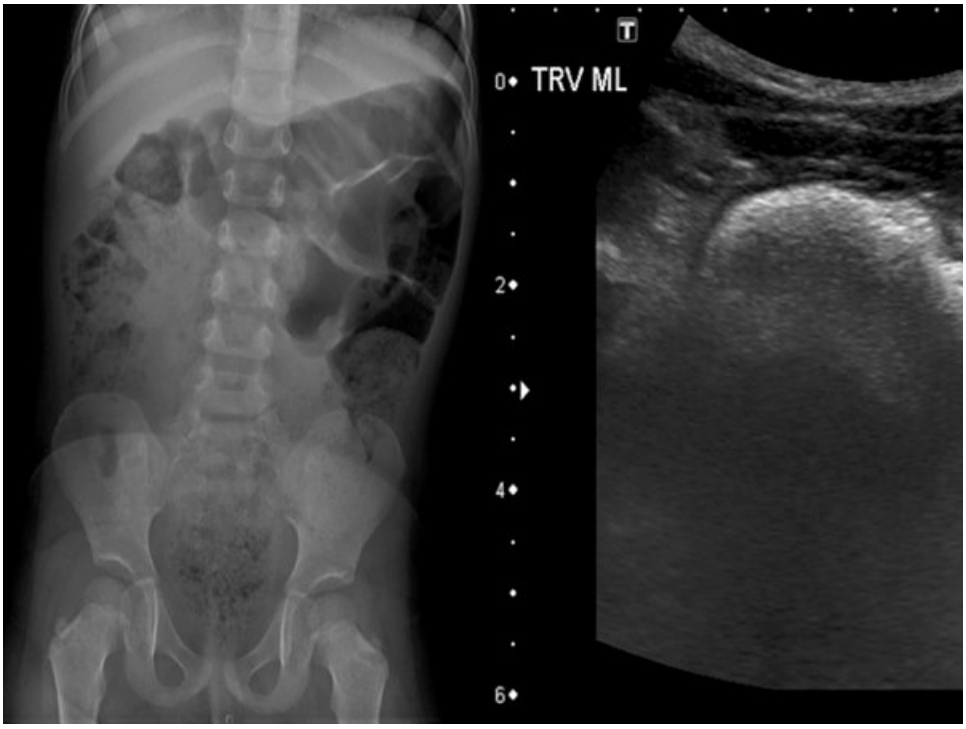


FIG. 36.1

4. **d. Digital rectal exam.** Physical exam should routinely include weight and height, and inspection of the perineum and genital and perianal regions (including anal position, stool present around the anus or on the underwear, signs of trauma, anal fissures, sensation). Although potentially considered to be an integral part of a complete physical exam, digital rectal examination should not be routinely conducted in children. It is reserved for difficult-to-treat cases and must be performed by health care professionals comfortable with interpreting features of anorectal anatomical abnormalities, to specifically evaluate for anal stenosis, a large fecal mass, or an empty rectum.
5. **e. Precocious puberty. Endocrine disorders associated with chronic dehydration (such as diabetes insipidus), electrolyte disorders (most notably hypercalcemia and hypokalemia), hypothyroidism, and hypervitaminosis D are important potential organic etiologies.** Precocious puberty presents with development of secondary sexual characteristics at an age before

the expected onset during normal development. Constipation is not a presenting or common isolated feature of this condition.

6. **c. Malone antegrade continence enema (MACE) channels and C-tubes provide better procedural independence than retrograde enemas for patients with neuropathic bowel dysfunction. In cases with severe constipation and a redundant colon, some have advocated the placement of conduits in the left colon rather than the cecum. By doing so, the length of bowel that has to be washed through is reduced and theoretically, so is the time taken for successful enema completion.** Results with this approach have been encouraging. Both open and laparoscopic procedures are associated with similar success rates. Both C-tubes and MACE-type channels provide independence and allow patients greater ability to perform irrigations without assistance in comparison to self-administered retrograde enemas. The underlying diagnosis influences the success rate. Patients with a neuropathic bowel and anorectal malformations seem to fare better than those with chronic idiopathic constipation. Age at operation is also important, with failures more commonly seen in younger patients irrespective of the diagnosis.
7. **a. Washout with regular evacuation of the entire colon.** The success of antegrade enema regimens is based on two important principles: (1) complete colonic emptying can achieve bowel continence, and (2) antegrade colonic emptying is feasible. Regular complete emptying of the colon is the main mechanism associated with fecal continence.
8. **a. Abdominal radiograph. Although of modest clinical value, abdominal radiographs are commonly used in the diagnosis and management of constipation. Proponents in favor of routine use argue that the study can clearly demonstrate the amount of fecal loading and delineate stool distribution throughout the colon and rectum, as well as help ascertain for the presence of fecal impaction. In addition, it may also reveal associated pathologies, such as bony abnormalities indicative of occult spinal dysraphism or sacral agenesis, and help provide a**

visual aid for family and patient recognition of stool retention despite a history of regular defecation. Colonic transit time studies are not recommended for routine diagnosis of functional constipation, being reserved for difficult-to-treat or unresponsive cases. Similarly, contrast enema series can be of value in selected cases, such as the evaluation of children with characteristics suggestive of Hirschsprung disease and repaired congenital anatomic abnormalities (i.e., anorectal malformation). Concern for a neuropathic process and/or lower spine stigmata should be evaluated with a spine ultrasound (if detected before calcification of the vertebral bodies in the first 3 to 6 months of life) or a lumbosacral MRI in older children. Anorectal manometry is useful only in very selected cases, such as suspected Hirschsprung disease and internal sphincter achalasia. In these conditions, the rectoanal relaxation reflex is absent. Nevertheless, in patients suspected of having functional constipation, manometry adds little to the diagnosis or therapeutic strategy.

9. **d. The appendix may be of sufficient length to be split to create a MACE and Mitrofanoff channel for neuropathic bowel and bladder management.** In children who require synchronous bladder reconstruction, a simultaneous MACE and Mitrofanoff urinary diversion offers the opportunity for dual fecal and urinary continence. If both a MACE and appendicovesicostomy are considered—and if the appendix is long enough with suitable, robust vascular anatomy—it is possible to split it. Previous surgical interventions are not an absolute contraindication for a laparoscopic approach, although the situation does demand for great care when entering the peritoneal cavity to avoid injuries related to adhesions or fibrosis. Although preoperative bowel preparation may facilitate the initiation of postoperative enemas, an aggressive cleanout is not necessary for the purpose of performing the procedure. Many patients with neuropathic bladder and bowel dysfunction have ventriculoperitoneal shunts. Although it is critical that appropriate use of prophylactic antibiotics and measures to minimize spillage be set in place, the presence of a shunt is not a contraindication for open or laparoscopic reconstruction involving bowel segments.

Many descriptions of the MACE procedure propose an "anti-reflux" valve mechanism to prevent leakage of bowel contents via the cutaneous stoma. This is often achieved by wrapping the appendix with the cecal wall. However, recent reports have suggested that it is not always necessary to construct an anti-reflux mechanism. Thus far, data appear to support no increase in stomal bowel incontinence, based on retrospective reviews comparing MACE with and without cecal wrap.

10. **a. It is a good alternative for patients who have previously undergone an appendectomy.** Cecostomy tubes are favored when the appendix is known to be absent (i.e., postappendectomy), when the patient refuses to perform intermittent bowel catheterization, for patients who develop stomal complications such as stenosis (as an alternative to revision of the MACE channel), as a temporary therapeutic challenge to determine response to antegrade enemas, to determine if ideal placement of a permanent MACE should be in the right or left colon, and as a permanent option in cases in which a nonoperative access is favored. The main drawback is that the tube entry site can become unsightly, with granulation tissue and occasional fecal leakage. Stenosis is very rare, in contrast to issues related to MACE channels. In addition, regular instrumentation is needed in order to change the tube on a scheduled basis (i.e., every 6 to 12 months), or sooner if it dislodges or breaks. Patients may opt for subsequent formal conversion to a bowel-based MACE, which can be done either laparoscopically or open. Compliance with antegrade enema regimens is crucial for success, irrespective of how the bowel access has been achieved (C-tube or MACE).
11. **d. Trial and error for more than 6 months may be warranted to reach a reliable enema routine. One of the most important points, especially in the early weeks and months after surgery, is to advise patients not to expect immediate success with the enema regimen, as early disappointment can lead to frustration and failure.** In fact many children may not achieve a steady state or a reliable enema routine for a period of as long as 6 months. Enema protocols differ among centers, and patients and families will frequently modify them to suit their own particular needs.

Initially, daily washouts with 20 mL/kg of solution are encouraged, but once the patient is comfortable with the process and a routine has been established, they may attempt to decrease frequency to alternate days. The time of day that the enema is administered is patient dependent, although most families prefer to give the enema during the early evening hours, after dinner. Purges can be done with large-volume tap or salt water, with the judicious mix of additives such as glycerin. The fluid does not have to be sterile.

12. **d. Prokinetic.** Prucalopride is a new oral, selective, high-affinity $5HT_4$ receptor antagonist with gastrointestinal prokinetic activities, which shows particular promise for management of difficult-to-treat constipation and may eventually represent a reasonable choice for children who fail to respond to more conservative measures. Its main mechanism of action does not influence stool consistency or bulk. As a prokinetic drug, it is likely to stimulate bowel smooth muscle contractions and not provide an antispasmodic effect.
13. **c. PEG.** With the introduction of PEG into routine clinical practice, tolerance of medical management has improved, and it is currently the preferred agent in many centers. **PEG is better tolerated and easier to administer than alternative medications such as lactulose, mineral oil, and milk of magnesia (magnesium hydroxide).** It is virtually tasteless and dissolves easily within seconds.
14. **a. Recurrence despite recommendations consistent with optimal medical management.** Functional constipation can be difficult to treat and a long-lasting problem for some children. Nevertheless, with adequate management close to 50% of patients monitored for 6 to 12 months can recover and successfully discontinue medications, whereas as many as 80% can be adequately controlled with routine interventions. Unfortunately, subsequent recurrences are fairly common, with as many as 50% of children experiencing one in the first 5 years after successful treatment. All the other listed factors (alternating constipation and diarrhea, episodes of bowel obstruction, bilious vomiting, and ribbon-like stools) are “warning signs and symptoms” that should raise suspicion for

alternative diagnosis and an underlying process (i.e., not functional constipation).

15. **d. Odor.** Stool characteristics should be recorded with a validated scale. The most commonly used (Bristol scale) takes into account consistency and stool shape, capturing also the degree of difficulty passing the bowel movement. Smell (odor) is not part of the scale.
16. **c. Rectal biopsy.** If Hirschsprung disease or colon aganglionosis is suspected, a deep suction rectal biopsy (including submucosal) should be obtained, favoring a transanal approach and aiming at a location 2 to 3 cm from the dentate line. Diagnosis is supported by absence of ganglion cells, hypertrophied nerve fibers, and increase in acetylcholinesterase activity in the lamina propria and muscularis mucosa.
17. **c. Presence of a large fecal load in the rectum and descending colon. Several difficulties can be experienced during enema infusion. The most common problem is pain or discomfort during instillation. In the majority of patients, this is a transient phenomenon that subsides during the first 3 months. It is always important to ensure that the pain is not due to distal fecal impaction, which can occur despite regular washouts.** The presence of a large amount of fecal material in the distal colon and rectum can certainly lead to impaction in patients doing infrequent antegrade enemas through a cecostomy access or MACE channel. Attempts at clearing this fecal load with antegrade flushes can lead to abdominal pain, lack of tolerance, and poor response. Spontaneous colon perforation with antegrade enema regimens is exceedingly rare. The use of hypotonic fluids (such as tap water) is commonplace in many centers. Retrograde flow of fluid into the distal ileum is not a common cause of pain with antegrade enemas and is unlikely to develop suddenly in a patient who has been doing enemas for a period of 6 months.
18. **a. Amount and distribution of fecal material can predict likelihood of response and recurrence with medical therapy.** As discussed in question 8, there are some potential benefits to obtaining an abdominal radiograph during the evaluation of children with constipation. These include assessment for fecal

impaction, to determine response to a bowel washout, to provide evidence for parents and caretakers, and to detect bony abnormalities suggestive of a possible neuropathic process. The distribution and amount of fecal material on one film have not been described to have any predictive value in terms of response to medical therapy or recurrence.

19. **e. All of the above.** Constipation is common and should be suspected in any patient who presents with lower urinary tract symptoms. Programs dealing with incontinence and dysfunctional voiding have successfully included this aspect of care into their protocols. Dysfunctional voiding and constipation should be addressed before proceeding with surgical correction and can be done by a pediatric urologist or urology nurse practitioner, assisted by the child's primary care physician.

Imaging

1. **d. Supplement fiber intake.** The physical examination and imaging studies are suggestive of severe constipation and stool impaction. Decreasing fecal load in the colon and rectum is the first step toward establishing an optimal medical regimen.**Impaction should be suspected when a mass is felt in the lower abdomen and/or left lower quadrant, or a dilated rectum filled with a large amount of stool is seen on pelvic ultrasound or abdominal radiography (as shown in Fig. 36.1**). Approximately 30% of children with functional constipation present with fecal impaction. Disimpaction and bowel washout address the problem in a relatively short period of time, often with enemas or suppositories, in contrast to maintenance therapy. Popular regimens also include "high-dose" PEG with or without sodium chloride, sodium phosphate, or mineral oil enemas. In some circumstances clearance can only be achieved with digital disimpaction under sedation or anesthesia. **The addition of fiber to the diet is bound to worsen the problem by increasing fecal load and is generally avoided during the initial management of fecal impaction.**

Chapter Review

1. Constipation may cause significant voiding dysfunction.
2. Functional constipation is a diagnosis of exclusion.
3. Organic conditions associated with elimination problems include cystic fibrosis, hypothyroidism, celiac disease, dietary allergies, Hirschsprung disease, anal stenosis, and trisomy 21. In the older child, mental health issues, eating disorders, sexual abuse, and irritable bowel syndrome should also be considered.
4. Initial medical management includes behavioral modification (regular defecation and nonsedentary activity), dietary changes (fluid and fiber intake), stool softeners and laxatives, and judicious use of enemas.
5. Antegrade continence enemas may be given through the cecum or left colon. It often requires that the child sit on the toilet for up to an hour before emptying is complete. The procedure is best employed in children 5–12 years of age who are motivated and is more successful in patients with neuropathic bowel or anorectal malformations.
6. Impaction should be suspected when a mass is felt in the lower abdomen and/or left lower quadrant, or a dilated rectum filled with a large amount of stool is seen on pelvic ultrasound or abdominal radiography.
7. In patients with fecal impaction, the addition of fiber to the diet is bound to worsen the problem by increasing fecal load and is generally avoided during the initial management.
8. The age of onset of symptoms is one of the easiest and most important pieces of information to obtain because it can be an important indicator for underlying pathology, particularly if it has been present since early in life (infancy).
9. Endocrine disorders associated with chronic dehydration (such as diabetes insipidus), electrolyte disorders (most notably hypercalcemia and hypokalemia), hypothyroidism, and hypervitaminosis D are important potential organic etiologies.

10. Although of modest clinical value, abdominal radiographs are commonly employed in the diagnosis and management of constipation. Proponents of routine use argue that the study can clearly demonstrate the amount of fecal loading and delineate stool distribution throughout the colon and rectum, as well as help ascertain the presence of fecal impaction. In addition, it may also reveal associated pathologies, such as bony abnormalities indicative of occult spinal dysraphism or sacral agenesis.
11. Polyethylene glycol is better tolerated and easier to administer than alternative medications such as lactulose, mineral oil, and milk of magnesia.
12. The most common problem with antegrade continence enemas is pain or discomfort during instillation. In the majority of patients, this is a transient phenomenon that subsides during the first 3 months. It is always important to ensure that the pain is not due to distal fecal impaction.

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Lower Urinary Tract Reconstruction in Children

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Keywords

augmentation; bladder neck channel; continent urinary diversion; reconstruction; urinary tract; urodynamics

Questions

1. Children with significant bladder or sphincter dysfunction requiring reconstructive surgery most likely have:
    a. bladder exstrophy or epispadias.
    b. posterior urethral valves.
    c. cloacal anomalies.
    d. prune-belly syndrome.
    e. spinal dysraphism.
2. The most important contribution to the field of pediatric reconstructive surgery has been:
    a. Mitrofanoff's description of a continent abdominal wall stoma using appendix.
    b. Lapides' introduction of clean intermittent catheterization (CIC).
    c. Goodwin's description of ileal reconfiguration.
    d. development of several effective means to increase bladder outlet resistance.

e. recognition that a dilated ureter could be used for bladder augmentation.

3. Normal bladder compliance is based on:
    a. ample collagen type II.
    b. inverse relationship of bladder volume and bladder pressure.
    c. bladder unfolding, elasticity, and viscoelasticity.
    d. subepithelial matrix bridges associated with collagen.
    e. hypertrophic bladder bundles interspersed with collagen.
4. Chronically elevated bladder filling pressures may cause hydronephrosis, vesicoureteral reflux, and impaired renal function. The lowest pressure threshold most often reported to cause problems is:
    a. 20 cm H_2O.
    b. 30 cm H_2O.
    c. 40 cm H_2O.
    d. 50 cm H_2O.
    e. 60 cm H_2O.
5. Upper urinary tract changes associated with a poorly compliant, hyperreflexic bladder are initially treated by:
    a. autoaugmentation.
    b. pharmacologic management and intermittent catheterization.
    c. ileal augmentation.
    d. sigmoid augmentation.
    e. gastric augmentation.
6. Preoperative bladder capacity and compliance are best determined by urodynamics using:
    a. carbon dioxide as an irrigant at a slow fill rate (10% of capacity per minute).
    b. room-temperature saline at a slow fill rate (10% of capacity per minute).
    c. body-temperature saline at a fast fill rate (30% of capacity per minute).
    d. cooled saline at a slow fill rate (10% of capacity per minute).
    e. cooled saline at a fast fill rate (30% of capacity per minute).
7. Urinary tract reconstruction for urinary continence requires:

a. confirmation of a normal upper urinary tract.
b. identification of a highly compliant bladder.
c. documentation of the presence or absence of vesicoureteral reflux.
d. acceptance of and compliance with intermittent catheterization.
e. documentation of a serum creatinine value less than 1.4 mg/dL.

8. Mechanical bowel preparation is performed in patients undergoing:
   a. ileocystoplasty.
   b. sigmoid cystoplasty.
   c. gastrocystoplasty.
   d. ureterocystoplasty.
   e. all of the above.
9. A urinary stricture after transureteroureterostomy is most likely due to:
   a. mobilization of the crossing ureter with periureteral tissue.
   b. mobilization of the crossing ureter without angulation beneath the inferior mesenteric artery.
   c. mobilization of the recipient ureter to meet the crossing one.
   d. wide anastomosis of the crossing ureter to the posteromedial aspect of the recipient.
   e. watertight anastomosis.
10. Creating an antireflux mechanism is most difficult with anastomosis to the:
   a. stomach.
   b. ileum.
   c. cecum.
   d. transverse colon.
   e. sigmoid colon.
11. The Young-Dees-Leadbetter bladder neck repair in children with neurogenic sphincter deficiency:
   a. results in limited success because of a lack of muscle tone and activity of the native bladder neck.
   b. can achieve successful continence results similar to those noted in children with bladder exstrophy.

c. does not often require bladder augmentation or intermittent catheterization.
d. is best performed in association with a Silastic sling.
e. limits the necessity for intermittent catheterization in children who could empty by a Valsalva maneuver preoperatively.

12. An ambulatory 15-year-old girl with lumbosacral myelomeningocele voids to completion with a low-pressure detrusor contraction and the Valsalva maneuver. She remains incontinent because of bladder neck and intrinsic sphincter deficiency that is refractory to pharmacologic management. To limit the risk of intermittent catheterization, the next step is:
    a. Young-Dees-Leadbetter bladder neck repair.
    b. artificial urinary sphincter placement.
    c. fascial bladder neck sling placement.
    d. Kropp bladder neck repair.
    e. Pippi-Salle bladder neck repair.
13. One side effect associated with bladder neck repair that can be decreased with good preoperative evaluation is:
    a. recurrent urolithiasis.
    b. recurrent cystitis.
    c. inability to spontaneously void.
    d. associated need for augmentation cystoplasty.
    e. unmasking of detrusor hostility, resulting in upper urinary tract changes.
14. Fascial slings used for increasing outlet resistance in children with neurogenic sphincteric incompetence:
    a. are more effective in girls than in boys.
    b. are dependent on the type of fascial or cadaveric tissue used.
    c. are dependent on the configuration of the sling and wrap used.
    d. rarely result in the need for bladder augmentation and intermittent catheterization.
    e. frequently result in urethral erosion.
15. The least favorable indication for an artificial urinary sphincter is:
    a. neurogenic bladder dysfunction.
    b. bladder exstrophy or epispadias.
    c. inability to empty the bladder by spontaneous voiding.

d. associated need for bladder augmentation.
e. prepubertal age.

16. The most common limitation of a Kropp urethral lengthening for continence is:
    a. fistula from the urethra to the bladder, resulting in incontinence.
    b. inability to spontaneously void, resulting in urinary retention.
    c. difficulty with intermittent catheterization, particularly in boys.
    d. new vesicoureteral reflux.
    e. distal ureteral obstruction.
17. Urinary continence is most definitively achieved after:
    a. Young-Dees-Leadbetter bladder repair.
    b. placement of an artificial urinary sphincter.
    c. placement of a circumferential fascial wrap.
    d. urethral lengthening and reimplantation.
    e. bladder neck division.
18. To avoid uninhibited pressure contractions during an enterocystoplasty:
    a. large bowel should be used.
    b. the intestinal segment should be reconfigured.
    c. the majority of the diseased bladder should be excised.
    d. a stellate incision into the bladder should be created to increase the circumference of the bowel anastomosis.
    e. small mesenteric windows are created in the bowel segment.
19. Potential ways to prevent reflux when using ileum for continent diversion include all of the following except:
    a. intussuscepted nipple valve.
    b. split nipple cuff of ureter.
    c. placement of the spatulated ureter into an incised mucosal trough.
    d. flap valve created beneath a taenia.
    e. placement of the ureter within a serosa-lined tunnel between two limbs of ileum.
20. The gastrointestinal segment that most often causes permanent gastrointestinal side effects when used in children with a

neurogenic bladder is the:

a. stomach.
b. jejunum.
c. ileum.
d. ileocecal segment.
e. sigmoid colon.

21. The most likely problem after gastrointestinal bladder augmentation is:

a. early satiety.
b. hyperchloremic metabolic acidosis.
c. small bowel obstruction.
d. chronic diarrhea.
e. vitamin B_{12} deficiency with megaloblastic anemia.

22. The gastrointestinal segment resulting in the best long-term capacity and compliance after augmentation cystoplasty is the:

a. gastric body.
b. gastric antrum.
c. ileum.
d. cecum.
e. sigmoid colon.

23. The risk of failure to achieve appropriate capacity and compliance after augmentation cystoplasty is:

a. less than 5%.
b. 5% to 10%.
c. 11% to 15%.
d. 16% to 20%.
e. more than 20%.

24. The serum metabolic pattern that occurs most often after an ileocystoplasty or colocystoplasty is:

a. hypochloremic metabolic acidosis.
b. hyperchloremic metabolic acidosis.
c. hypochloremic metabolic alkalosis.
d. hyperchloremic metabolic alkalosis.
e. hyponatremic metabolic acidosis.

25. The serum metabolic pattern that occurs most often after gastrocystoplasty is:
    a. hypochloremic metabolic acidosis.
    b. hyperchloremic metabolic acidosis.
    c. hypochloremic metabolic alkalosis.
    d. hyperchloremic metabolic alkalosis.
    e. hyponatremic metabolic acidosis.
26. The risk of intermittent hematuria and dysuria after gastrocystoplasty is most influenced by:
    a. the gastric segment used.
    b. persistent urinary incontinence.
    c. decreased renal function.
    d. diagnosis of bladder exstrophy.
    e. neurogenic bladder dysfunction.
27. Bacteriuria should be treated after bladder augmentation when:
    a. associated with CIC.
    b. urinalysis demonstrates microscopic hematuria.
    c. there is increased mucus production.
    d. etiology is posterior urethral valves.
    e. urine culture reveals growth of a urea-splitting organism.
28. The gastrointestinal segment associated with the lowest incidence of stone formation is:
    a. stomach.
    b. jejunum.
    c. ileum.
    d. cecum.
    e. sigmoid colon.
29. Adenocarcinoma of the bladder after augmentation cystoplasty can occur after:
    a. 2 years.
    b. 4 years.
    c. 8 years.
    d. 16 years.
    e. 26 years.
30. The risk of perforation after bladder augmentation includes all but:
    a. high outflow resistance.

b. persistent hyperreflexia or uninhibited bladder contractions.
c. use of sigmoid colon.
d. bladder exstrophy.
e. neurogenic bladder dysfunction.

31. The initial management of a spontaneous perforation of an augmented bladder in a child with a neurogenic bladder is:
    a. placement of a large-bore urethral catheter for drainage.
    b. placement of a large-bore suprapubic cystotomy tube for drainage.
    c. immediate surgical exploration and repair.
    d. serial abdominal examinations.
    e. urine culture.
32. Pregnancy associated with urinary reconstruction:
    a. is reasonable after urinary diversion but is contraindicated after augmentation cystoplasty.
    b. results in the mesenteric pedicle positioned directly anterior to the uterus.
    c. results in the mesenteric pedicle deflected laterally without vascular compromise to the augmented segment.
    d. is avoided due to mechanical compression of the pedicle and ischemia with loss of the augmented segment.
    e. is contraindicated because of increased risk of systemic sepsis complicating the hydronephrosis.
33. Ureterocystoplasty is limited because:
    a. it requires an intraperitoneal approach.
    b. complete mobilization of the ureter may result in vascular compromise.
    c. a dilated ureter is not as compliant as a similar-sized bowel segment.
    d. a dilated ureter is not available in many patients.
    e. ureterocystoplasty precludes spontaneous voiding.
34. Autoaugmentation is contraindicated with:
    a. serum creatinine value greater than 1.4 ng/dL.
    b. CIC.
    c. vesicoureteral reflux.
    d. uninhibited bladder contractions.

e. small bladder capacity.

35. A ureterosigmoidostomy should not be undertaken in a patient with a history of:
    a. dilated ureters.
    b. anteriorly placed rectum associated with bladder exstrophy.
    c. recurrent pyelonephritis.
    d. fecal incontinence.
    e. constipation.
36. The use of efferent nipple valves for continence in children:
    a. has not approached the results achieved in adults.
    b. has a higher complication and reoperation rate than a flap valve.
    c. is equivalent to any other continence mechanism.
    d. is often associated with difficulty in catheterization.
    e. often results in stomal stenosis.
37. The least important factor when creating an appendicovesicostomy is:
    a. taking a wide cecal cuff to decrease the risk of stomal stenosis.
    b. creating a tunnel of 4 cm, at least greater than a 5:1 ratio of tunnel length to diameter, to achieve continence.
    c. a small, uniform lumen allowing for easy catheterization.
    d. mobilizing the right colon to adequately free the appendix.
    e. tubularizing a small portion of the cecum in continuity with the appendix to increase length.
38. A frequent occurrence after an appendicovesicostomy is:
    a. urinary incontinence due to inadequate length of the flap valve mechanism.
    b. urinary incontinence due to persistently elevated reservoir pressure.
    c. appendiceal perforation that often occurs due to catheterization.
    d. appendiceal stricture or necrosis.
    e. stomal stenosis.
39. A 12-year-old obese girl with spina bifida undergoes appendicocecostomy, bladder neck sling, bladder augmentation,

and continent catheterizable bladder channel. The upper urinary tract is normal. The best source of tissue for the bladder channel is:

a. distal right ureter after right-to-left transureteroureterostomy.
b. tapered segment of small bowel of adequate length.
c. right fallopian tube.
d. gastric tube.
e. tubularized bladder flap.

40. In complex pediatric urinary undiversion procedures it is most difficult to:

a. provide adequate outflow resistance.
b. create a compliant urinary reservoir.
c. achieve an effective antireflux mechanism without upper tract obstruction.
d. provide a reliable access for intermittent catheterization.
e. achieve urinary and fecal continence.

Answers

1. **e. Spinal dysraphism.** Most pediatric reconstructive procedures are undertaken to correct a problem of the native urinary tract causing progressive hydronephrosis, urinary incontinence unresponsive to medical management, or temporary diversion. Children with bladder and sphincteric dysfunction are the most complex reconstructive cases seen in pediatric urology; children with the diagnoses of exstrophy, persistent cloaca and urogenital sinus, posterior urethral valves, bilateral single ectopic ureters, and prune-belly syndrome may be involved. However, children with a neurogenic bladder due to a myelomeningocele make up the vast majority of patients requiring this type of surgical intervention.
2. **b. Lapides' introduction of clean intermittent catheterization (CIC).** One of the most important contributions in the care of children with bladder dysfunction came with the acceptance of CIC described by Lapides and colleagues in 1972 and 1976, based on the work of Guttmann and Frankel. The effective use of CIC has allowed the application of augmentation and lower tract

reconstruction to groups of patients who had not previously been candidates. The principle of intermittent catheterization allows the reconstructive surgeon to aggressively correct storage problems by providing an adequate reservoir and good outflow resistance. Spontaneous voiding, although a goal, is not imperative because catheterization can be used for emptying.

3. **c. Bladder unfolding, elasticity, and viscoelasticity.** Multiple factors contribute to the property of compliance. Initially the bladder is in a collapsed state, which allows for the storage of urine at low pressure by simply unfolding. While it expands, detrusor properties of elasticity and viscoelasticity take effect. Elasticity allows the detrusor muscle to stretch without an increase in tension until it reaches a critical volume. When filling is slow, as in a natural state, or stops, there is a rapid decay in this pressure known as stress relaxation. Normally, stress relaxation is in balance with the filling rate and prevents an increase in detrusor pressure.
4. **c. 40 cm H_2O. Elevated passive filling pressure becomes clinically pathogenic when a pressure greater than 40 cm H_2O is chronically reached.** Pressure at this level sustained during a prolonged period of time impairs ureteral drainage and can result in acquired vesicoureteral reflux, pyelocaliceal changes, hydroureteronephrosis, and decreased glomerular filtration rate.
5. **b. Pharmacologic management and intermittent catheterization.** Pharmacologic management can play a role in decreasing filling pressure, particularly when hyperreflexic detrusor contractions are present. A combination of medications and intermittent catheterization has a positive impact, particularly in children with neurogenic dysfunction.
6. **b. Room-temperature saline at a slow fill rate (10% of capacity per minute).** The testing medium and infusion rate can influence the results. Carbon dioxide is not as reliable as fluid infusion, particularly when evaluating bladder compliance and capacity. The most common fluids used for testing are saline and iodinated contrast material; both provide reproducible results. Use of testing media at body temperature is also appropriate, but room

temperature has also been shown to be acceptable. End filling pressure and bladder compliance can be dramatically affected by simply changing the filling rate. **The cystometrogram should be performed at a fill rate of 10% per minute of the predicted bladder capacity for age.**

7. **d. Acceptance of and compliance with intermittent catheterization.** No test ensures that a patient will be able to void spontaneously and empty well after bladder augmentation or other reconstruction. **Therefore, all patients must be prepared to perform CIC postoperatively.** The native urethra should be examined for the ease of catheterization. Ideally, the patient should learn CIC and practice it preoperatively until the patient, family, and surgeon are comfortable that catheterization can and will be performed reliably. In spite of a technically perfect operation, failure to catheterize and empty the bladder after reconstruction can result in upper tract deterioration, urinary tract infection, or bladder perforation.
8. **e. All of the above.** Each patient undergoes preoperative bowel preparation to minimize the potential risk of surgery if the use of any bowel is contemplated. Even when ureterocystoplasty or other alternatives are planned, intraoperative findings may dictate the need for use of a bowel segment.
9. **c. Mobilization of the recipient ureter to meet the crossing one.** If the native urinary bladder is small and adequate for only a single ureteral tunnel, transureteroureterostomy and a single reimplant may be helpful. **Typically, the better ureter should be implanted into the bladder. The contralateral ureter drains into the reimplanted ureter via a transureteroureterostomy.** The crossing ureter should follow a smooth path and remain tension free. It should be carefully mobilized with all of its adventitia and as much periureteral tissue as possible to preserve blood supply. Care must be taken not to angulate the crossing ureter beneath the inferior mesenteric artery. The crossing ureter should be widely anastomosed to the posteromedial aspect of the recipient ureter. **The recipient ureter should not be mobilized or brought**

medially to meet the contralateral ureter to minimize devascularization.

10. **b. Ileum.** The necessity of ureteral reimplantation into an intestinal segment may occasionally determine the segment to be used for bladder augmentation or replacement. Long-term experience with ureterosigmoidostomy and colon conduit diversion has established an effective means of creating a nonrefluxing ureteral implant. If a gastric segment is used for bladder augmentation or replacement, the ureters may be implanted into the stomach in a manner remarkably similar to that used in the native bladder. Creating an effective antireflux mechanism into an ileal segment is more difficult. The split nipple technique described by Griffith may prevent reflux at least at low reservoir pressure.
11. **a. Results in limited success because of a lack of muscle tone and activity of the native bladder neck.** Reports of success with the Young-Dees-Leadbetter bladder neck reconstruction in children with neurogenic sphincter dysfunction are limited, not only in the number of series but also in overall improvement of incontinence. **Independent reviews of long-term results of this repair show minimal success in individuals with neurogenic dysfunction.** These authors speculate that the lack of success was due to a lack of muscle tone and activity in the wrapped muscle related to the neurogenic problem.
12. **b. Artificial urinary sphincter placement.** The artificial urinary sphincter has been recognized as the only procedure that can result in prompt continence in selected children while preserving their ability to void spontaneously.
13. **e. Unmasking of detrusor hostility, resulting in upper urinary tract changes. It is now recognized that occlusion of the bladder neck in children with neurogenic sphincter incompetence can result in the unmasking or development of detrusor hostility manifest by a decrease in bladder compliance or increase in detrusor hyperreflexia.** Careful preoperative urodynamic assessment helps to identify some of the children who are at risk.
14. **a. Are more effective in girls than in boys.** Fascial slings have been used more extensively and with better results in girls with

neurogenic sphincter incompetence, although recently some success has been reported in boys. Overall long-term success with fascial slings in the neurogenic population has varied greatly from 40% to 100%.

15. **c. Inability to empty the bladder by spontaneous voiding. The ultimate benefits of the artificial urinary sphincter include its ability to achieve a high rate of continence while maintaining the potential for spontaneous voiding.** For practical purposes, when intermittent catheterization is required along with augmentation cystoplasty, using native tissue for continence eliminates the long-term concern for infection/erosion and the risk of mechanical failure.
16. **c. Difficulty with intermittent catheterization, particularly in boys.** One study examined the results in 23 children, 22 of whom had neurogenic sphincter incompetence, and noted continence in more than 90% of the children. The most common complication was difficult catheterization, particularly in boys. Fewer than half of the boys in this series were catheterized through the native urethra; the majority were catheterized via an abdominal wall stoma.
17. **e. Bladder neck division.** The ultimate procedure to increase bladder outlet resistance is to divide the bladder neck so that it is no longer in continuity with the urethra. This must be accompanied by creation of a continent abdominal wall stoma and should be performed only in patients who will reliably be able to perform catheterization.
18. **b. The intestinal segment should be reconfigured.** Two studies demonstrated the advantages of opening a bowel segment on its antimesenteric border, which allows detubularization and reconfiguration of that intestinal segment. Reconfiguration into a spherical shape provides multiple advantages, including maximization of the volume achieved for any given surface area, blunting of bowel contractions, and improvement of overall capacity and compliance.
19. **d. Flap valve created beneath a taenia.** Small bowel does not have a taenia; this method is appropriate for large bowel. The split

nipple technique described by Griffith may prevent reflux at least at low reservoir pressure. LeDuc and colleagues in 1987 described a technique in which the ureter is brought through a hiatus in the ileal wall. From that hiatus the ileal mucosa is incised and the edges are mobilized so as to create a trough for the ureter. It may also be possible to create antireflux mechanism using a serosal-lined tunnel created between two limbs of ileum as described by Abol-Enein and Ghoneim in 1999. Reinforced nipple valves of ileum have been used extensively to prevent reflux with the Kock pouch. Good long-term results have been achieved by Skinner after several modifications.

20. **d. Ileocecal segment.** Chronic diarrhea after bladder augmentation alone is rare. Diarrhea can occur after removal of large segments of ileum from the gastrointestinal tract, although the length of the segments typically used for augmentation is rarely problematic unless other problems coexist. Removal of the ileum and ileocecal valve from the gastrointestinal tract may cause diarrhea. One study noted that 10% of patients with neurogenic dysfunction have significant diarrhea after such displacement.
21. **b. Hyperchloremic metabolic acidosis.** Postoperative bowel obstruction is uncommon after augmentation cystoplasty, occurring in approximately 3% of patients after augmentation. The rate of obstruction is equivalent to that noted after conduit diversion or continent urinary diversion. Removal of the distal ileum from the gastrointestinal tract may result in vitamin B_{12} deficiency and megaloblastic anemia. The terminal 15 to 20 cm of ileum should not be used for augmentation, although problems may arise even if that segment is preserved. Early satiety may occur after gastrocystoplasty but usually resolves with time. Disorders of gastric emptying should be extremely rare, particularly when using the body of the stomach.
22. **c. Ileum. Ileal reservoirs have been noted to have lower basal pressures and less motor activity when created for continent urinary diversion.** Problems with pressure after augmentation cystoplasty usually occur from uninhibited contractions caused by

the bowel segment. It is extremely rare not to achieve an adequate capacity or flat tonus limb unless a technical error has occurred with use of the bowel segment. Rhythmic contractions have been noted postoperatively with all bowel segments, particularly the stomach, although ileum is the least likely to demonstrate a remarkable urodynamic abnormality.

23. **b. 5% to 10%.** Hollensbe and associates at Indiana University reported on one of the largest experiences with pediatric bladder augmentation and found that approximately 5% of patients had significant uninhibited contractions causing clinical problems. Another study found that 6% required secondary augmentation of a previously augmented bladder for similar problems in long-term follow-up.
24. **b. Hyperchloremic metabolic acidosis.** The first recognized metabolic complication related to storage of urine within intestinal segments was the occasional development of hyperchloremic metabolic acidosis after ureterosigmoidostomy. Another study demonstrated the mechanisms by which acid is absorbed from urine in contact with intestinal mucosa. **A later report noted that essentially every patient after augmentation with an intestinal segment had an increase in serum chloride and a decrease in serum bicarbonate levels, although clinically significant acidosis was rare if renal function was normal.**
25. **c. Hypochloremic metabolic alkalosis.** Gastric mucosa is a barrier to chloride and acid resorption and, in fact, secretes hydrochloric acid. The secretory nature of gastric mucosa may at times be detrimental to the patient and can result in two unique complications of gastrocystoplasty. **Severe episodes of hypokalemic hypochloremic metabolic alkalosis after acute gastrointestinal illnesses have been noted after gastrocystoplasty.**
26. **e. Neurogenic bladder dysfunction.** Virtually all patients with normal sensation have occasional hematuria or dysuria with voiding or catheterization after gastrocystoplasty beyond that which is expected with other intestinal segments. All patients should be warned of this potential problem, although in most

patients these symptoms are intermittent and mild and do not require treatment. The dysuria is less problematic in patients with limited sensation due to neurogenic dysfunction. Patients who are incontinent or have decreased renal function may be at increased risk. These problems occur less frequently after antral gastric cystoplasty in which there is a smaller load of parietal cells.

27. **e. Urine culture reveals growth of a urea-splitting organism. It appears that the use of CIC is a prominent factor in the development of bacteriuria in patients after augmentation cystoplasty. Every episode of asymptomatic bacteriuria does not require treatment in patients performing CIC. Bacteriuria should be treated when significant symptoms occur, such as fever, suprapubic pain, incontinence, and gross hematuria.** Bacteriuria should also be treated when the urine culture demonstrates growth of a urea-splitting organism that may lead to stone formation.
28. **a. Stomach.** Most bladder stones in the augmented child are of a struvite composition. Bacteriuria has been thought to be an important risk factor. Stones have been noted after the use of all intestinal segments with no significant difference appreciated between small and large intestine. Struvite stones are less likely after gastrocystoplasty.
29. **b. 4 years.** Patients undergoing augmentation cystoplasty should be made aware of a potential increased risk of tumor development. Yearly surveillance of the augmented bladder with endoscopy should eventually be performed; the latency period until such procedures are necessary is not well defined. The earliest reported tumor after augmentation was found only 4 years after cystoplasty.
30. **d. Bladder exstrophy.** The cause of delayed perforations after bladder augmentation is unknown. Perforations may occur in bladders with significant uninhibited contractions after augmentation. High outflow resistance may maintain bladder pressure rather than allowing urinary leakage and venting of the pressure, potentially increasing ischemia. **The majority of patients suffering perforations after augmentation cystoplasty have a neurogenic etiology.** At Indiana University, perforations were

noted in 32 of 330 patients undergoing cystoplasty an average of 4.3 years after augmentation. Analysis of this experience suggested that the use of sigmoid colon was the only significant increased risk.

31. **c. Immediate surgical exploration and repair.** The standard treatment of spontaneous perforation of the augmented bladder is surgical repair, as it is for intraperitoneal rupture of the bladder after trauma. **The majority of patients with perforations have myelodysplasia and present late in the course of the disease because of impaired sensation.** Increasing sepsis and death of the patient may result from a delay in diagnosis or treatment.
32. **c. Results in the mesenteric pedicle deflected laterally without vascular compromise to the augmented segment.** Experience is limited regarding what is known about the changes to the pedicle of a bladder augmentation during pregnancy. It has been reported that the mesenteric pedicle to bladder augmentations is not stretched over the uterus at the time of cesarean section. The pedicle has been found to be deflected laterally. Urinary tract infections may be problematic in women who have undergone urinary reconstruction, including bladder augmentation. Ureteral dilatation, increased residual urine, and diminished tone to the upper tract may all be important risk factors.
33. **d. A dilated ureter is not available in many patients.** Several series have reported good results after ureteral augmentation with a follow-up as long as 8 years. The upper urinary tract has remained stable or improved in virtually all patients. Complications are uncommon. The main disadvantage to ureterocystoplasty is the limited patient population with a poorly functioning kidney drained by a megaureter.
34. **e. Small bladder capacity. Although autoaugmentation can improve compliance, an increase in volume is "modest at best."** In a report of 12 children who had undergone a detrusorotomy, five were considered to have excellent results, two had acceptable results, and one was lost to follow-up. The main disadvantage of autoaugmentation is a limited increase in bladder capacity such that

adequate preoperative volume may be the most important predictor of success.

35. **d. Fecal incontinence.** Before ureterosigmoidostomy is considered, anal sphincter competence must be ensured. Tests used to assess sphincter integrity include manometry, electromyography, and practical evaluation of the ability to retain an oatmeal enema in the upright position for a time period without soilage. Incontinence of a mixture of stool and urine results in foul soilage and must be avoided.
36. **b. Has a higher complication and reoperation rate than a flap valve.** The greatest experience with nipple valves for achieving urinary continence has been with the Kock pouch. Skinner and associates made a series of modifications to aid in maintenance of the efferent nipple. Even with experience and these modifications, a failure rate of 15% or higher can be expected. Equivalent results with the nipple valve and a Kock pouch have been achieved in children.
37. **b. Creating a tunnel of 4 cm, at least greater than a 5:1 ratio of tunnel length to diameter, to achieve continence.** The appendix is an ideal natural tubular structure that can be safely removed from the gastrointestinal tract without significant morbidity. The small caliber of the appendix facilitates creation of a short functional tunnel with the bladder wall. **Experience has shown that continence can be achieved with only a 2-cm appendiceal tunnel.**
38. **e. Stomal stenosis.** Incontinence is rare with the Mitrofanoff procedure and may result from inadequate length of the flap valve mechanism or persistently elevated reservoir pressure. **The most common complication has been stomal stenosis and occurs in 10% to 20% of patients.** Stenosis resulting in difficult catheterization may occur early in the postoperative course and requires formal revision.
39. **b. Tapered segment of small bowel of adequate length.** When the appendix is unavailable for use, other tubular structures can provide a similar mechanism for catheterization and continence. Mitrofanoff, in 1980, described a similar technique using ureter.

Woodhouse and MacNeily, in 1994, as well as others, have used the fallopian tube, which can accommodate catheterization. Monti and Yang have been credited with a novel modification of the tapered intestinal segment, which can be reimplanted according to the Mitrofanoff principle.

40. **c. Achieve an effective antireflux mechanism without upper tract obstruction.** The key to urinary undiversion is understanding the original pathologic condition that led to diversion. One report described a 26-year experience with urinary undiversion in 216 patients. In that series, management of the bladder was relatively straightforward and effective with bladder augmentation as necessary. Inadequate outflow resistance was usually treated with Young-Dees-Ledbetter bladder neck repair. **Most complications were related to the ureters; 23 patients required reoperation for persistent reflux, whereas 10 did so for partial obstruction of the ureter.** Those reoperation rates are indicative of the difficulty one faces in dealing with short, dilated, and scarred ureters, which may be present after urinary diversion.

Chapter Re view

1. Bladder volume (mL) in children is equal to 30 × (age in years + 2) or for infants, 7 × weight (kg).
2. Intermittent catheterization must be taught and accepted by the patient and caregiver before any urinary reconstruction is performed.
3. There is no test that ensures the patient will be able to void spontaneously and empty well after bladder augmentation or reconstruction.
4. Most patients prefer to catheterize an abdominal wall stoma rather than the native urethra.
5. Bladder neck bulking agents are not particularly effective in children.
6. When placing an artificial sphincter it should be placed at the bladder neck in females and in prepubertal males.

7. One-third of patients will require further surgery after augmentation cystoplasty because of various problems.
8. Bacteriuria is common after intestinal cystoplasty. After intestinal cystoplasty, routine bladder irrigation should be performed to evacuate inspissated mucus.
9. The use of stomach for bladder augmentation should be reserved for patients who have short-gut syndrome or who have received heavy pelvic irradiation.
10. Delayed spontaneous perforation of the bowel segment after intestinal cystoplasty occurs in approximately 5% of patients.
11. Most secondary vesicoureteral reflux will resolve after successful bladder reconstruction.
12. Nonfunctional bladders may need to be cycled to determine their true capacity.
13. Removing the ileal-cecal valve from the gastrointestinal tract in patients with neurogenic bladder and bowel dysfunction may result in intractable diarrhea.
14. It has been noted that there appears to be an increased incidence of malignant tumors in the gastric segment of patients who have had a gastrocystoplasty.
15. When the appendix is used to create a flap valve, the distance to the skin should be as short as possible to facilitate ease of catheterization.
16. The Young-Dees-Leadbetter bladder neck reconstruction in children with neurogenic sphincter dysfunction has had limited success.
17. Occlusion of the bladder neck in children with neurogenic sphincter incompetence can result in the unmasking or development of detrusor hostility manifest by a decrease in bladder compliance or increase in detrusor hyperreflexia.
18. Ileal reservoirs have been noted to have lower basal pressures and less motor activity when created for continent urinary diversion.
19. Essentially every patient after augmentation with an ileal or colonic intestinal segment has an increase in serum chloride and a decrease in serum bicarbonate levels, although severe acidosis is rare if renal function is normal.

20. Severe episodes of hypokalemic hypochloremic metabolic alkalosis after acute gastrointestinal illnesses have been noted after gastrocystoplasty.
21. The majority of patients suffering bladder perforations after augmentation cystoplasty have a neurogenic etiology.